

General- p Toeplitz Kernels: A Dual-Distance Criterion and a Minimal-Function Reduction

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Lane 0 — candidate partial result

11 August 2026

Status. Substantial partial answer. For every $1 < p < \infty$ we give an exact functional-analytic criterion for a Toeplitz kernel to consist only of bounded functions. We also prove that the minimal-function question has a positive answer for every Toeplitz kernel possessing an exact outer-multiplier model-space representation. What remains is the structural, non-Hilbertian factorisation problem needed to make that representation universal.

1 The two source questions

Câmara–Malheiro–Partington first characterize those upper-half-plane model spaces K_θ^p contained in H_+^∞ . They then ask for a complete answer for general Toeplitz kernels in H_+^p , outside the known Hilbert case.

Theorem 4.3. $K_\theta^p \subset H_\infty^+$ if and only if

$$\sup_{w \in \mathbb{C}^+} \frac{1 - |\theta(w)|^2}{\operatorname{Im} w} < \infty.$$

Proof. Note that, by Dyakonov's result, it is sufficient to discuss the case $p = 2$. By the closed graph theorem a necessary and sufficient condition for K_θ^2 to embed into H_∞^+ is that, for all $f \in K_\theta^2$, we have $f \in H_\infty^+$ and there is a constant $C > 0$ such that

$$\|f\|_\infty \leq C\|f\|_2 \quad (4.7)$$

for all $f \in K_\theta^2$.

Since for $f \in K_\theta^2$ we have $\sup_{w \in \mathbb{C}^+} |f(w)| = \sup_{w \in \mathbb{C}^+} |\langle f, k_w^\theta \rangle|$, condition (4.7) is equivalent to the condition that the L_2 norms of the k_w^θ are uniformly bounded, independently of w . For $p = 2$ we have

$$\|k_w^\theta\|_2^2 = \langle k_w^\theta, k_w^\theta \rangle = |k_w^\theta(w)|$$

and the result follows from (3.7). $\square$

The following refinement of (4.5) is an immediate consequence of Theorem 4.3 and (4.5) itself.

Corollary 4.4. We have $K_\theta^p \subset H_\infty^+$ if and only if

$$\lim_{\varepsilon \rightarrow 0^+} \inf\{|\theta(z)| : 0 < \operatorname{Im} z < \varepsilon\} = 1. \quad (4.8)$$

Dyakonov's condition that $\inf\{|\theta(z)| : 0 < \operatorname{Im} z < \varepsilon\} > 0$ for some $\varepsilon > 0$ has appeared elsewhere in the literature, being applied to realization theory [15] and finite-time controllability [14]. (The context is the right half-plane but it is easy to transcribe the results for the upper half-plane.) In particular, for a Blaschke product with zeroes $\lambda_n = x_n + iy_n$, $n \geq 1$, the condition is shown in [15] to be equivalent to the property that $\inf y_n > 0$ and

$$\sup_{x \in \mathbb{R}} \sum_{n=1}^{\infty} \frac{y_n}{y_n^2 + (x - x_n)^2} < \infty,$$

which in turn can be expressed as a Carleson measure condition on the measure $\mu := \sum_{n=1}^{\infty} y_n \delta_{\lambda_n}$, tested on reproducing kernels k_λ lying on a horizontal line.

A more general question, to which we do not know a complete answer except in the case $p = 2$, is to ask when a T-kernel contains only bounded functions.

Figure 1: Source PDF page 14: the general boundedness question.

The second question uses the following terminology. If $\mathcal{K} = \mathcal{K}_{\min}(IO)$ for an inner I and outer O , then O is a *minimal function* for $\mathcal{K}$ when $\mathcal{K}_{\min}(O) = \operatorname{span}\{O\}$. The source proves existence for all model spaces and asks whether every nonzero proper Toeplitz kernel has such a function.

where $I_+ \prec I_+ I_{1+}$. This cannot happen, however, when $\ker T_{\overline{O}_+/O_+} = \text{span}\{O_+\}$, which is equivalent to saying that O_+^2 is rigid in $H_{p/2}^+$ ([2]). In fact, (5.12) would imply that $I_+ I_{1+} O_{1+} \in \ker T_{I_+ \overline{O}_+/O_+}$ and thus $I_{1+} O_{1+} \in \ker T_{\overline{O}_+/O_+} = \text{span}\{O_+\}$, which is impossible for non-constant θ_1 .

Definition 5.7. *If $g \in L_\infty$, we say that O_+ is a minimal function for $\ker T_g$ if and only if for some inner function I_+ we have $\ker T_g = \mathcal{K}_{\min}(I_+ O_+)$ and $\mathcal{K}_{\min}(O_+) = \text{span}\{O_+\}$.*

In H_2^+ , every non-trivial T-kernel has a minimal function ([18],[19]). The following theorem shows that this property also holds for model spaces in H_p^+ ; whether the same is true in general for T-kernels in H_p^+ is an open question, to the authors' knowledge.

Theorem 5.8. *For any $p \in (1, \infty)$ and any inner function θ , there exists a minimal function O_+ in K_θ .*

Proof. With the notation of (3.20) and (5.7), it is enough to consider $O_+ = \frac{h_+}{\lambda_{+0}^+}$ and $I_+ = B_{\lambda_{+0}^+}^{\lambda_{+0}^+}$. $\square$

6 On the relations between $\ker T_g$ and $\ker T_{\theta g}$

If θ is a non-constant inner function, $g \in L_\infty$ and $\ker T_g \neq \{0\}$, we have $\ker T_{\theta g} \subsetneq \ker T_g$. We may then ask how much “smaller” $\ker T_{\theta g}$ is, with respect to $\ker T_g$, and in particular when is it non-trivial.

Definition 6.1. *Let $g \in L_\infty$ and θ be an inner function. If $\ker T_g \neq \{0\}$ and $\ker T_{\theta g} = \{0\}$, we say that θ annihilates $\ker T_g$.*

It is clear that a necessary and sufficient condition for $\ker T_g$ not to be annihilated by θ is that there exists φ_+ such that

$$\theta \varphi_+ \in \ker T_g, \quad \varphi_+ \in H_p^+ \setminus \{0\}, \quad (6.1)$$

and in this case $\varphi_+ \in \ker T_{\theta g}$.

If θ is a finite Blaschke product we have the following result from [1], taking into account that in this case $\theta \sim r^k$, where k is the number of zeroes of θ .

Figure 2: Source PDF page 19: Definition 5.7 and the general minimal-function question.

2 An exact general-p boundedness criterion

Work first on the upper half-plane, exactly as in the source. Put $q = p/(p-1)$ and use the duality

$$\langle f, h \rangle = \frac{1}{2\pi} \int_{\mathbb{R}} f(x) \overline{h(x)} dx \quad (f \in H_+^p, h \in H_+^q).$$

For $w \in \mathbb{C}_+$ let $k_w(x) = (x - \overline{w})^{-1}$. Cauchy's formula gives $|f(w)| = |\langle f, k_w \rangle|$. For $g \in L^\infty(\mathbb{R})$ define

$$\mathcal{R}_g := \overline{T_g(H_+^q)}^{H_+^q}.$$

Theorem 1 (adjoint-range distance criterion). *Let $1 < p < \infty$, $g \in L^\infty(\mathbb{R})$, and $\mathcal{K} = \ker(T_g : H_+^p \rightarrow H_+^p)$. Then*

$$\mathcal{K} \subset H_+^\infty \iff \sup_{w \in \mathbb{C}_+} \text{dist}_{H_+^q}(k_w, \mathcal{R}_g) < \infty. \quad (2.1)$$

In fact the supremum in (2.1) is exactly the operator norm of the inclusion $\mathcal{K} \hookrightarrow H_+^\infty$ under the normalization above. If the supremum is infinite, $\mathcal{K}$ contains an unbounded function.

Proof. The adjoint of T_g under $H_+^p - H_+^q$ duality is $T_{\bar{g}}$. The annihilator identity for a bounded operator therefore gives

$$\mathcal{K}^\perp = (\ker T_g)^\perp = \overline{\text{ran } T_g^*} = \mathcal{R}_g.$$

By Hahn–Banach, restriction identifies $\mathcal{K}^*$ isometrically with $H_+^q/\mathcal{K}^\perp$. Hence

$$\|\text{ev}_w|_{\mathcal{K}}\| = \|k_w + \mathcal{K}^\perp\|_{H_+^q/\mathcal{K}^\perp} = \text{dist}_{H_+^q}(k_w, \mathcal{R}_g). \quad (2.2)$$

Taking the supremum first over w and then over the unit ball of $\mathcal{K}$ shows that the right side of (2.1) is precisely the norm of the inclusion, provided that inclusion is bounded.

It remains only to justify that set inclusion implies bounded inclusion. If $f_n \rightarrow f$ in $\mathcal{K}$ with the H^p norm and $f_n \rightarrow F$ in the H^∞ norm, continuity of point evaluations in H^p and locally uniform convergence from H^∞ give $f(w) = F(w)$ for every w . Thus the graph is closed, and the closed graph theorem applies. The last assertion is the contrapositive: if every member were bounded, the closed graph argument would make the supremum finite. $\square$

This is an exact answer in quotient geometry, valid without closed-range or Fredholm hypotheses. It is deliberately not advertised as the intrinsic symbol-factorisation classification sought by the source: the closure $\mathcal{R}_g$ can itself be difficult to compute.

3 What an exact model-space representation buys

For the second question it is cleaner to use the conformally equivalent disk notation. Let $K_\theta^p = \ker T_\theta \subset H^p(\mathbb{D})$ and assume $\theta(0) = 0$. Recall that

$$\mathcal{K}_{\min}(f) = \ker T_{\bar{z}\bar{I}\bar{O}/O} \quad \text{when } f = IO$$

is the inner–outer factorization of f . The standard Cayley isometry gives the corresponding upper-half-plane statement.

Theorem 2 (outer model multipliers are minimal functions). *Let $1 < p < \infty$. Suppose a nonzero proper Toeplitz kernel $\mathcal{K} \subset H^p(\mathbb{D})$ has an exact representation*

$$\mathcal{K} = wK_\theta^p, \quad (3.1)$$

where θ is inner with $\theta(0) = 0$ and w is outer. Then

$$\mathcal{K}_{\min}(w) = \text{span}\{w\}, \quad (3.2)$$

$w\theta/z$ is a maximal function for $\mathcal{K}$, and therefore w is a minimal function for $\mathcal{K}$ in the sense of the source.

Proof. Since $1 \in K_\theta^p$, (3.1) gives $w \in \mathcal{K}$. Minimality of $\mathcal{K}_{\min}(w)$ among Toeplitz kernels containing w gives $\mathcal{K}_{\min}(w) \subset \mathcal{K}$. If $f \in \mathcal{K}_{\min}(w)$, write $f = wk$ with $k \in K_\theta^p \subset H^p$. On the other hand, $f \in \ker T_{\bar{z}\bar{w}/w}$ implies

$$\frac{f}{w} = \frac{\bar{h}}{\bar{w}} \quad \text{for some } h \in H^p.$$

Thus f/w belongs both to $H^p \subset \mathcal{N}^+$ and to $\overline{\mathcal{N}^+}$. The Smirnov intersection theorem makes it constant. This proves (3.2).

Now θ/z is inner and is a maximal function for K_θ^p . Let $m = w\theta/z$ and suppose $m \in \ker T_a$ for a bounded symbol a . Every $k \in K_\theta^p$ can be written $k = (\theta/z)\bar{n}$ with $n \in \mathcal{N}^+$ and with the displayed product in H^p . Hence every $f = wk \in \mathcal{K}$ has the form $f = m\bar{n}$. Since $am \in \overline{H_0^p}$ and $af \in L^p \cap \overline{\mathcal{N}^+}$, the Smirnov theorem gives $af \in \overline{H_0^p}$. Therefore $\mathcal{K} \subset \ker T_a$, proving $\mathcal{K}_{\min}(m) = \mathcal{K}$. The inner–outer factors of m are θ/z and w , and (3.2) is exactly the remaining clause in the definition of a minimal function. $\square$

Corollary 3 (boundedness inside the same class). *Under (3.1), put $q = p/(p - 1)$ and $k_\lambda(z) = (1 - \bar{\lambda}z)^{-1}$. Then*

$$\mathcal{K} \subset H^\infty \iff \sup_{\lambda \in \mathbb{D}} |w(\lambda)| \operatorname{dist}_{H^q}(k_\lambda, \theta H^q) < \infty. \quad (3.3)$$

For $p = 2$, the distance in (3.3) is $\sqrt{(1 - |\theta(\lambda)|^2)/(1 - |\lambda|^2)}$.

Proof. Apply Theorem 1, in disk form, to K_θ^p and then to the evaluation of wk . Since $(K_\theta^p)^\perp = \theta H^q$, the norm of evaluation at λ on K_θ^p is the displayed distance. Multiplication by w multiplies that evaluation functional by $w(\lambda)$. The $p = 2$ formula is the norm of the usual model-space reproducing kernel. $\square$

4 Why the result remains partial

In H^2 , Hayashi’s theorem represents every nontrivial Toeplitz kernel as wK_θ^2 with an outer isometric multiplier w ; Theorem 2 then recovers the positive Hilbert answer. Away from $p = 2$, the strongest general representation found in the literature is the extremal/intersection form recorded by O’Loughlin (reformulating Hartmann–Seip): for an extremal q_0 and inner I ,

$$\ker T_g = q_0 K_I^p \cap H^p \quad (p \leq 2), \quad \ker T_g = q_0 K_I^2 \cap H^p \quad (p \geq 2). \quad (4.1)$$

The intersection in (4.1) cannot be removed unless multiplication by q_0 maps the entire indicated model space into H^p . That missing onto-divisor property is precisely what is automatic in the Hilbert case and what would turn Theorem 2 into a full answer.

Six focused upgrade attempts were made: adjoint-range duality, closed-range factorisation, extremal-vector rigidity, exact model multipliers, removal of the intersection in (4.1), and transfer across $p = 2$. The last route fails in opposite directions: for $p > 2$ a Hilbert minimal factor need not lie in H^p , while for $p < 2$ rigidity in H^1 need not imply rigidity in the larger $H^{p/2}$ class. The packet therefore stops at the strongest proved claims rather than assuming the non-Hilbert divisor theorem it would need.

5 Novelty and verification

The two proofs above are self-contained apart from standard Hardy-space duality, the Smirnov intersection theorem, and the source’s minimal-kernel formula. Searches through the exact phrases, later arXiv literature, and the run indexes did not locate these two statements verbatim. The distance criterion is an elementary Banach-duality consequence and should be treated as a useful reformulation unless expert review establishes novelty. The model-multiplier theorem is the mathematically stronger partial advance.

References

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- [2] R. O’Loughlin, *Toeplitz kernels and the backward shift*, J. Math. Anal. Appl. 492 (2020), 124489, [arXiv:2001.10890](#).
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- [4] M. C. Câmara, C. Carteiro, and C. Diogo, *Maximal functions, conjugations and multipliers between Toeplitz kernels*, [arXiv:2512.20406](#).